



Vidyavardhaka Sangha®, Mysore
VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi
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Department of Computer Science & Engineering

Phone: +91 821-4276230, Email: hodcs@vvce.ac.in

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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-1

COURSE NAME : SYSTEM SOFTWARE & COMPILER DESIGN		ACADEMIC YEAR : 2023-2024		
COURSE CODE : 21CS63		DATE : 21-05-2024		
SEM / BRANCH : VI/CSE		MAX. MARKS : 15		
FACULTY : Dr. Aditya C R, Dr. Chethana H T, Prof. Lakshmi B S		DURATION : 1 Hr		
Q No.	Instructions to students: Answer all the Questions	Marks Allotted	BL	CO
MODULE 1				
1 a)	Explain the different phases of a compiler with translation of statement "Sum = Var X 10" at each phase. Assume Sum and Var are of type float.	07	L2	CO1
1 b)	Construct transition diagrams to recognize the following tokens. i) Unsigned Numbers ii) Identifiers	08	L3	CO2

Course Outcomes:

CO1	:	Explain the basic concepts of system software and compiler design
CO2	:	Apply the concepts of compiler design and assembler features for designing system software
CO3	:	Analyze and trace the working of lexical analyzer and syntax analyzer for a given grammar
CO4	:	Design and develop solutions to system software problems using LEX and YACC in a team (PO5 and PO9)



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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-2

COURSE NAME : SYSTEM SOFTWARE & COMPILER DESIGN	ACADEMIC YEAR : 2023-2024
COURSE CODE : 21CS63	DATE : 20-06-2024
SEM / BRANCH : VI/CSE	MAX. MARKS : 30
FACULTY : Dr. Aditya C R, Dr. Chethana H T, Prof. Lakshmi B S	DURATION : 1 Hr

Q No.	Instructions to students: Answer one full question from each module	Marks Allotted	BL	CO
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MODULE 2

1 a)	Construct first and follow set for the given grammar. $E \rightarrow E+T \mid T$ $T \rightarrow T * F \mid F$ $F \rightarrow (E) \mid id$	08	L3	CO2
1 b)	Verify whether the given grammars are suitable for non-recursive predictive parsing. If not, state the reasons and make the grammars suitable for top-down parsing. i) $A \rightarrow 01A10 \mid 0110$ ii) $B \rightarrow Ba+ \mid Bc* \mid a$	07	L4	CO3

OR

2 a)	Terminal /Input	num	(	+	)	\$	08	L3	CO2
	S	$S \rightarrow ES'$	$S \rightarrow ES'$						
	S'			$S' \rightarrow +S$	$S' \rightarrow \epsilon$	$S' \rightarrow \epsilon$			
	E	$E \rightarrow a$	$E \rightarrow (S)$						
Using the above parse table, parse the string “ (a+a) + a ”									
2 b)	Verify whether any conflicts arise for the given string (a,(a,a)) in SR parser using the grammar.						07	L4	CO3
	$S \rightarrow (L) \mid a$ $L \rightarrow L,S \mid S$								

MODULE 3

3 a)	Explain Synthesized Attribute and Inherited Attribute.	04	L2	CO1																
3 b)	Construct LR(0) automaton for the given grammar S -> AA A -> aA b	07	L3	CO2																
3 c)	For the given syntax directed definition (SDD), verify the dependency between the variables in the expression 3 * 5 + 4 using S-attributed SDD and construct topological ordering. <table><tr><th>PRODUCTION</th><th>SEMANTIC RULES</th></tr><tr><td>1) L → E n</td><td>L.val = E.val</td></tr><tr><td>2) E → E₁ + T</td><td>E.val = E₁.val + T.val</td></tr><tr><td>3) E → T</td><td>E.val = T.val</td></tr><tr><td>4) T → T₁ * F</td><td>T.val = T₁.val × F.val</td></tr><tr><td>5) T → F</td><td>T.val = F.val</td></tr><tr><td>6) F → (E)</td><td>F.val = E.val</td></tr><tr><td>7) F → digit</td><td>F.val = digit.lexval</td></tr></table>	PRODUCTION	SEMANTIC RULES	1) L → E n	L.val = E.val	2) E → E ₁ + T	E.val = E ₁ .val + T.val	3) E → T	E.val = T.val	4) T → T ₁ * F	T.val = T ₁ .val × F.val	5) T → F	T.val = F.val	6) F → (E)	F.val = E.val	7) F → digit	F.val = digit.lexval	04	L4	CO3
PRODUCTION	SEMANTIC RULES																			
1) L → E n	L.val = E.val																			
2) E → E ₁ + T	E.val = E ₁ .val + T.val																			
3) E → T	E.val = T.val																			
4) T → T ₁ * F	T.val = T ₁ .val × F.val																			
5) T → F	T.val = F.val																			
6) F → (E)	F.val = E.val																			
7) F → digit	F.val = digit.lexval																			

OR



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4 a)

Explain Syntax directed definition (SDD) with an example.

04

L2

CO1

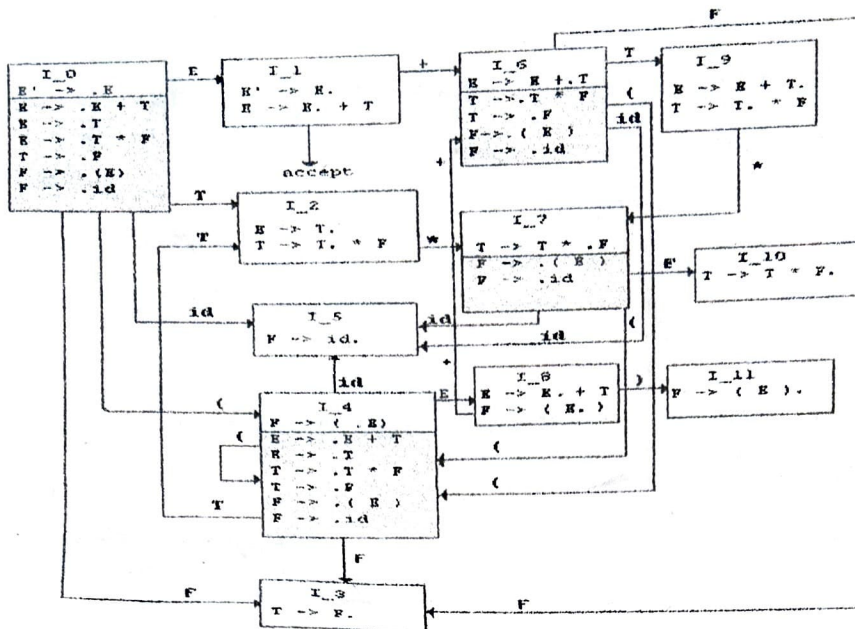
4 b)

Construct goto and action table for the given LR(0) automaton.

07

L3

CO2



Given the grammar –

$E \rightarrow E + T \mid T$

$T \rightarrow T * F \mid F$

$F \rightarrow (E) \mid id$

4 c)

For the given syntax directed definition (SDD), verify the dependency between the variables in the expression $3 * 5$ using L attributed SDD and construct topological ordering.

04

L4

CO3

PRODUCTION	SEMANTIC RULES
1) $T \rightarrow F T'$	$T'.inh = F.val$ $T.val = T'.syn$
2) $T' \rightarrow * F T'_1$	$T'_1.inh = T'.inh \times F.val$ $T'.syn = T'_1.syn$
3) $T' \rightarrow \epsilon$	$T'.syn = T'.inh$
4) $F \rightarrow digit$	$F.val = digit.lexval$

Course Outcomes:

CO1	:	Explain the basic concepts of system software and compiler design
CO2	:	Apply the concepts of compiler design and assembler features for designing system software
CO3	:	Analyze and trace the working of lexical analyzer and syntax analyzer for a given grammar
CO4	:	Design and develop solutions to system software problems using LEX and YACC in a team (PO5 and PO9)



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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-3

COURSE NAME : SYSTEM SOFTWARE & COMPILER DESIGN	ACADEMIC YEAR : 2023-2024
COURSE CODE : 21CS63	DATE : 19-07-2024
SEM / BRANCH : VI/CSE	MAX. MARKS : 30
FACULTY : Dr. Aditya C R, Dr. Chethana H T, Prof. Lakshmi B S	DURATION : 1 Hr

Q No.	Instructions to students: Answer all the questions	Marks Allotted	BL	CO
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MODULE 4

1 a)	Explain the issues in the design of code generator.	07	L2	CO1
1 b)	Generate assembly code for the below three address code using three registers. Given p, q, r and s are actual variables and t, u, v are temporaries. t = p - q u = p - r v = t + u p = s s = v + u	08	L3	CO2

MODULE 5

2 a)	Explain the data structures used in assembler.	05	L2	CO1
2 b)	Generate the object program for the following source statement of SIC by generating the object code for each statement.	10	L3	CO2

	COPY	START	5000
5000	FIRST	STL 14	RETADR
5003	CLOOP	JSUB 48	RDREC
5006		LDA 60	LENGTH
5009		COMP 28	ZERO
5010		JEQ 30	ENDFIL
5013		JSUB	WRREC
5012		J 30	CLOOP
5015	ENDFIL	LDA	EOF
5018		STA 00	BUFFER
5016		LDA	THREE
5010		STA	LENGTH
5021		JSUB	WRREC
5024		LDL 08	RETADR
5023		RSUB 40	
5025	EOF	BYTE	C'EOF'
5024	THREE	WORD	3
5021	ZERO	WORD	0
5020	RETADR	RESW	1
5021	LENGTH	RESW	1
5020	BUFFER	RESB	4096

END

FIRST



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Given: STL=14, JSUB=48, LDA = 00, COMP= 28, JEQ=30, J=3C,
STA=0C, LDL=08, RDREC = 6039, WRREC = 6061, E=69, O=79 and
F=70

Course Outcomes:

C01	:	Explain the basic concepts of system software and compiler design
C02	:	Apply the concepts of compiler design and assembler features for designing system software
C03	:	Analyze and trace the working of lexical analyzer and syntax analyzer for a given grammar
C04	:	Design and develop solutions to system software problems using LEX and YACC in a team (PO5 and PO9)



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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-1

COURSE NAME :	Machine Learning	ACADEMIC YEAR :	2023-2024		
COURSE CODE :	21CS61	DATE :	20/05/2024		
SEM / BRANCH :	VI/CSE	MAX. MARKS :	15		
FACULTY :	Dr. Paramesha K, Dr. Vedavathi N, & Prof. Tanuka K	DURATION :	1 Hr.		
Q No.	Instructions to students: Answer the following questions	Marks Allotted	BL	CO	
MODULE 1					
1 a)	Explain the different types of data in machine learning with suitable examples.	8	L2	CO1	
1 b)	Analyze the following scenarios w.r.t to implementation of machine learning algorithms and answer the following questions: (i) Recommend suitable machine learning techniques (supervised, unsupervised and reinforcement learning algorithm). (ii) Justify and state the process briefly on how to implement the machine learning technique for the given task. SCENARIO 1: A researcher has utilized a web scraper to amass a collection of news articles spanning diverse topics, with the aim of constructing a corpus of documents. Subsequently, the researcher seeks to determine the quantity of documents amassed for each specific topic. What machine learning approach would be suitable for accomplishing this task? Justify? SCENARIO 2: In a manufacturing facility, an assembly line utilizes industrial robots to handle spare parts, lifting them onto and off the assembly line. These robotic arms operate in three-dimensional space, moving at specific speeds to reach and lift the spare parts. What machine learning techniques are employed to control the arm movements effectively and ensure the smooth operation of the plant?	7	L4	CO3	

COURSE OUTCOMES

CO1	Understand the fundamentals of machine learning
CO2	Illustrate the working of machine learning algorithms.
CO3	Analyze theory of probability, the concepts of neural network and classifier for problems in machine learning
CO4	Demonstrate machine learning models for predictive analytics.



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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-2

COURSE NAME : Machine Learning	ACADEMIC YEAR : 2023-2024
COURSE CODE : 21CS61	DATE : 19/06/2024
SEM / BRANCH : VI/CSE	MAX. MARKS : 30
FACULTY : Dr. Paramesha K, Dr. Vedavathi N, & Prof. Tanuja K	DURATION : 1 Hr.

Q No.	Instructions to students: Answer one full question from each module	Marks Allotted	BL	CO
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MODULE 2

Construct Decision tree to for the following dataset in Table 1.1 to predict Target Label.

Table 1.1

A1	A2	A3	LABEL
TRUE	HOT	HIGH	NO
TRUE	HOT	HIGH	NO
FALSE	HOT	HIGH	YES
FALSE	COOL	NORMAL	YES
FALSE	COOL	NORMAL	YES
TRUE	COOL	HIGH	NO
TRUE	HOT	HIGH	NO
TRUE	HOT	NORMAL	YES
FALSE	COOL	NORMAL	YES
FALSE	COOL	HIGH	NO

1 a)

8

L3

CO2

1 b)

Examine the reasons for overfitting in the decision tree model and provide solutions to reduce the same.

7

L4

CO3

OR

2 a)

Construct Decision tree for the following dataset in Table 2.1 to predict Target Label Poisonous

Table 2.1

Example	Colour	Toughness	Fungus	Appearance	Poisonous
1	Green	Soft	Yes	Wrinkled	Yes
2	Green	Hard	Yes	Smooth	No
3	Brown	Soft	No	Wrinkled	No
4	Brown	Soft	Yes	Wrinkled	Yes
5	Green	Soft	Yes	Smooth	Yes
6	Green	Hard	No	Wrinkled	No
7	Orange	Soft	Yes	Wrinkled	Yes

8

L3

CO2

2 b)

Examine the effects of continuous valued attributes, many valued attributes on modeling decision trees and suggest the solution to overcome the issues briefly.

7

L4

CO3

MODULE-3

3 a)

Discuss the appropriate problems for neural network learning and explain Gradient Descent algorithm for training a perceptron.

7

L2

CO1



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3 b)	Train a perceptron to realize function of AND gate with two inputs and compute weights in each iteration starting with initial weights $w_1=0.4$ and $w_2=0.5$, threshold of step function = 1 and learning rate = 0.5	8	L3	CO2
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029 1

COURSE OUTCOMES

CO1	Understand the fundamentals of machine learning
CO2	Illustrate the working of machine learning algorithms.
CO3	Analyze theory of probability, the concepts of neural network and classifier for problems in machine learning.
CO4	Demonstrate machine learning models for predictive analytics.

VIDYAVARDHAKA COLLEGE OF ENGINEERINGAutonomous Institute, Affiliated to Visvesvaraya Technological University, Belagavi
Gokulam 3rd Stage, Mysuru 570 002**Sixth Semester B.E. Examinations: August 2024**
Machine Learning

Duration: 3 Hour

Max. Mark: 100

INSTRUCTION TO STUDENTS

- 1) Answer **One Full** question from each module
- 2) Three module questions are compulsory and the remaining two modules will have internal choice

Q.No	Module-I	Marks	BL	CO
1(a)	Explain the detailed activities involved in building a machine learning model with a neat diagram.	10	L2	CO1
1(b)	Analyze the following scenario w.r.to implementation of machine learning algorithms and answer the following questions- (i) Recommend suitable machine learning techniques (supervised, unsupervised and reinforcement learning algorithm) (ii) Justify and state the process briefly on how to implement the machine learning technique for the task Scenario a. An email service provider wants to automatically filter out spam emails to improve user experience by ensuring that users receive only legitimate emails in their inboxes. b. Imagine in a class there are 50 students. Five project teams need to be created for a project work. Faculty in charge for the project work collects data of all the students and employs an automated ML based interface to carry out the team formation. What is the nature of ML technique suitable in this case.	10	L4	CO3

Module-II

- 2(a) Find the information gain from the following dataset by considering the discrete valued attributes only. (NOTE: Target Attribute=Decision)

Sl. No	Speed	Distance to intersection	Traffic Control Device	Pedestrian Presence	Decision
1	0.5	0.7	Stop sign	Pedestrian crossing	Slow down
2	0.2	0.9	Green traffic light	No pedestrian crossing	Stop
3	0.8	0.4	Red traffic light	No pedestrian crossing	Go
4	0.1	0.1	No traffic light	Pedestrian crossing	Slow down
5	0.5	0.5	No traffic sign	Pedestrian crossing	Go
6	0.8	0.8	No traffic sign	No pedestrian crossing	Stop
7	0.2	0.2	Green traffic light	No pedestrian crossing	Stop

8	0.6	0.6	Stop sign	No pedestrian crossing	Slow down
9	0.3	0.3	No traffic sign	No pedestrian crossing	Stop
10	0.5	0.5	Red traffic light	Pedestrian crossing	Go

- 2(b)** Analyze the following Boolean function and construct a decision tree for the same.
 (i) $A \wedge \neg B$
 (ii) $A \vee [B \wedge C]$
 (iii) $A \text{ XOR } B$
 (iv) $[A \wedge B] \vee [C \wedge D]$

10

L4

C03

OR

- 3(a)** Considering the following dataset, to create a decision tree that accurately predicts the suspension system's performance, which attribute is the most important amongst Road Condition and Suspension Type?

10

L3

C02

Sl. No	Road Condition	Suspension Type	Speed (mph)	Temperature (F)	Performance
1	Wet	Independent	50	70	Good
2	Dry	Dependent	70	80	Good
3	Wet	Independent	30	60	Poor
4	Dry	Dependent	60	90	Good
5	Wet	Independent	40	75	Poor
6	Dry	Dependent	80	85	Good
7	Wet	Independent	20	50	Poor
8	Dry	Dependent	50	70	Poor
9	Wet	Independent	60	80	Good
10	Dry	Dependent	40	65	Poor

- 3(b)** Analyze any three issues in decision tree learning with example.

10

L4

C03

Module-III

- 4(a)** Explain the backpropagation algorithm for feedforward networks.

10

L2

C01

- 4(b)** Implement the perceptron learning rule for AND gate using step function with bias/ $w_0 = -0.4$, $w_1=0.3$, $w_2 = -0.2$ and learning rate $=0.3$, using the step function gives output 1 if $f(x)$ is greater than 0 else it outputs 0.

10

L3

C02

Module-IV

- 5(a)** Apply the Naïve Bayes classifier to classify the new instance $X = (\text{age} = \text{youth}, \text{income} = \text{medium}, \text{student} = \text{yes}, \text{credit rating} = \text{fair})$ either yes or no class using the below given dataset, AllElectronics Customer Database

10

L3

C02

Table 1: AllElectronics Computer Database

RID	Age	Income	Student	Credit_rating	Class: buys comp
1.	Youth	High	No	Fair	No
2.	Youth	High	No	Excellent	No
3.	Middle-aged	High	No	Fair	Yes
4.	Senior	Medium	No	Fair	Yes
5.	Senior	Low	Yes	Fair	Yes

	6.	Senior	Low	Yes	Excellent	No				
	7.	Middle-aged	Low	Yes	Excellent	Yes				
	8.	Youth	Medium	No	Fair	No				
	9.	Youth	Low	Yes	Fair	Yes				
	10.	Senior	Medium	Yes	Fair	Yes				
	11.	Youth	Medium	Yes	Excellent	Yes				
	12.	Middle-aged	Medium	No	Excellent	Yes				
	13.	Middle-aged	High	Yes	Fair	Yes				
	14.	Senior	Medium	No	Excellent	No				

5(b)	Analyze Brute Force MAP concept learning for consistent and inconsistent hypothesis.	10	L4	CO3
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OR

6(a)	Consider a football game between two rival teams: Team 0 and Team 1. Suppose Team 0 wins 95% of the time and Team 1 wins the remaining matches. Among the games won by team 0, only 30% of them come from playing on team 1's football field. On the other hand, 75% of the victories for team 1 are obtained while playing at home. If team 1 is to host the next match between the two teams, which team will most likely emerge as the winner?	10	L3	CO2
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6(b)	Analyze Minimum Description Length Principle for choosing hypothesis to minimize sum of two description lengths.	10	L4	CO3
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Module-V

7(a)	Explain Q learning algorithm with an example.	10	L2	CO1
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7(b)	Apply KNN classifier to predict the weight of a person with the given features Height and Age, with K=3 and K=5 for a given new instance in the below dataset.	10	L3	CO2
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ID	HEIGHT	AGE	WEIGHT
1	5	45	77
2	5.11	26	47
3	5.6	30	55
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
7	5.3	19	40
8	5.8	28	60
9	5.5	23	45
10	5.6	32	58
11	5.5	38	?

----End----

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Gokulam 3rd Stage, Mysuru 570 002

Seventh Semester B.E. Examinations: December 2024 – January 2025

Blockchain Technology

Duration: 3 Hour

Max. Mark: 100

INSTRUCTION TO STUDENTS

- 1) Answer **One Full** question from each module.
- 2) Three module questions are compulsory and remaining two modules will have internal choice.

Q.No	Module-I	Marks	BL	CO/PO
1(a)	Define Blockchain. Explain the key features of blockchain.	8	L2	CO1/PO1
1(b)	Differentiate between Proof of Work and Proof of Stake.	6	L2	CO1/PO1
1(c)	Explain the difference applications of Blockchain technology.	6	L2	CO1/PO1
OR				
2(a)	What are the requirements which must be met to provide the desired results in a consensus mechanism? Discuss the same.	6	L2	CO1/PO1
2(b)	Illustrate the Generic Elements of blockchain.	8	L2	CO1/PO1
2(c)	"CAP theorem has been violated in Blockchain". Justify the statement.	6	L2	CO1/PO1
Module-II				
3(a)	Describe the following with respect to blockchain architecture. (i) Centralized System (ii) Decentralized system	4	L2	CO1/PO1
3(b)	Illustrate the various modes of operations for the following Block ciphers. (i) Electronic Code book (ii) Cipher Block chaining	8	L3	CO2/PO1
3(c)	Illustrate Generic cryptography models with a neat diagram.	8	L3	CO2/PO1
Module-III				
4(a)	Illustrate the working principle of SHA algorithm.	10	L3	CO2/PO1
4(b)	Analyze RSA algorithm for a given two prime numbers 7 and 13. Generate public key and private key and encrypt the message "WELCOME".	10	L4	CO3/PO2
Module-IV				
5(a)	Define smart contract. Enumerate the properties of smart contract.	6	L2	CO1/PO1
5(b)	Analyze how the given instruction is executed. Using P2PKH script execution. <sig><pubk> op_Dup op_HASH160<PubHash> OP_EQUAL VERIFY OP_CHECKSIG	7	L4	CO3/PO2
5(c)	Analyze the properties of Ricardian contract.	7	L4	CO3/PO2
Module-V				
6(a)	Illustrate the following (i) Nonce (ii) Gasprice (iii) Data (iv) Init	7	L3	CO2/PO1
6(b)	Illustrate the concept of GAS in Ethereum	7	L3	CO2/PO1
6(c)	Analyze the relationship between transaction, transaction trie and block header.	6	L4	CO3/PO2
OR				
7(a)	Illustrate the working of ethereum virtual machine with diagram.	7	L3	CO2/PO1
7(b)	Illustrate different types of Accounts in ethereum Blockchain.	7	L3	CO2/PO1
7(c)	Analyze the process of mining to validate and verify the blocks.	6	L4	CO3/PO2

---End---



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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-2

COURSE NAME : Blockchain Technology	ACADEMIC YEAR : 2024-2025
COURSE CODE : 21CS724	DATE : 25/11/2024
SEM / BRANCH : VII/CSE	MAX. MARKS : 30
FACULTY : Lakshmi B S	DURATION : 1 Hr.

Q No.	Instructions to students: Answer one full question from each module	Marks Allotted	BL	CO/PO Mapping
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MODULE 3

1 a)	Illustrate Merkle tree with an example	3	L3	CO2& PO1
1 b)	Illustrate Hash Function with its 3 security properties	7	L3	CO2& PO1
1 c)	Consider two prime numbers 3 and 11, Calculate the public and private keys and encrypt the message "HELLO"	5	L4	CO3& PO2

MODULE 4

2 a)	Explain Bitcoin with keys and addresses	3	L2	CO1 & PO1
2 b)	Consider the scenario "Sender A is transferring 100 bitcoins to receiver B along with 128-bit keys", Analyze by using transaction life cycle.	12	L4	CO3& PO2

COURSE OUTCOMES

CO1	Understand the blockchain terminologies with its applications.
CO2	Examine and apply the decentralization, asymmetric cryptographic primitives, and Bitcoin concepts
CO3	Analyze the principles of Ethereum and its transactions in blockchain
CO4	Design and demonstrate a blockchain using Ethereum Tool and make effective presentation in a team



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CONTINUOUS COMPREHENSIVE EVALUATION: TEST-3

COURSE NAME : Blockchain Technology	ACADEMIC YEAR : 2024-2025
COURSE CODE : 21CS724	DATE : 12/12/2024
SEM / BRANCH : VII/CSE	MAX. MARKS : 15
FACULTY : Lakshmi B S	DURATION : 1 Hr.

Q No.	Instructions to students: Answer one full question	Marks Allotted	BL	CO/PO Mapping
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MODULE- 5

1 a)	Illustrate the working principle of Ethereum virtual machine with a neat diagram.	8	L3	CO2& PO1
1 b)	Examine the following Data and draw the state transition function capturing the essence of transaction. Data "Sender Address is 4718bf7a and Recipient Address is 74F7A2 and sender sends 2 Ethers to recipient".	7	L4	CO3& PO2
[OR]				
2 a)	Consider an example of Ethereum transaction and write transaction using common transaction fields.	8	L3	CO2& PO1
2 b)	Analyze the given Data and evaluate the total gas cost for a transaction in Ethereum blockchain. Data "SHA3 has gas cost of 30 and current gas price is 25 Gwei"	7	L4	CO3& PO2

COURSE OUTCOMES

CO1	Understand the blockchain terminologies with its applications.
CO2	Examine and apply the decentralization, asymmetric cryptographic primitives, and Bitcoin concepts
CO3	Analyze the principles of Ethereum and its transactions in blockchain
CO4	Design and demonstrate a blockchain using Ethereum Tool and make effective presentation in a team